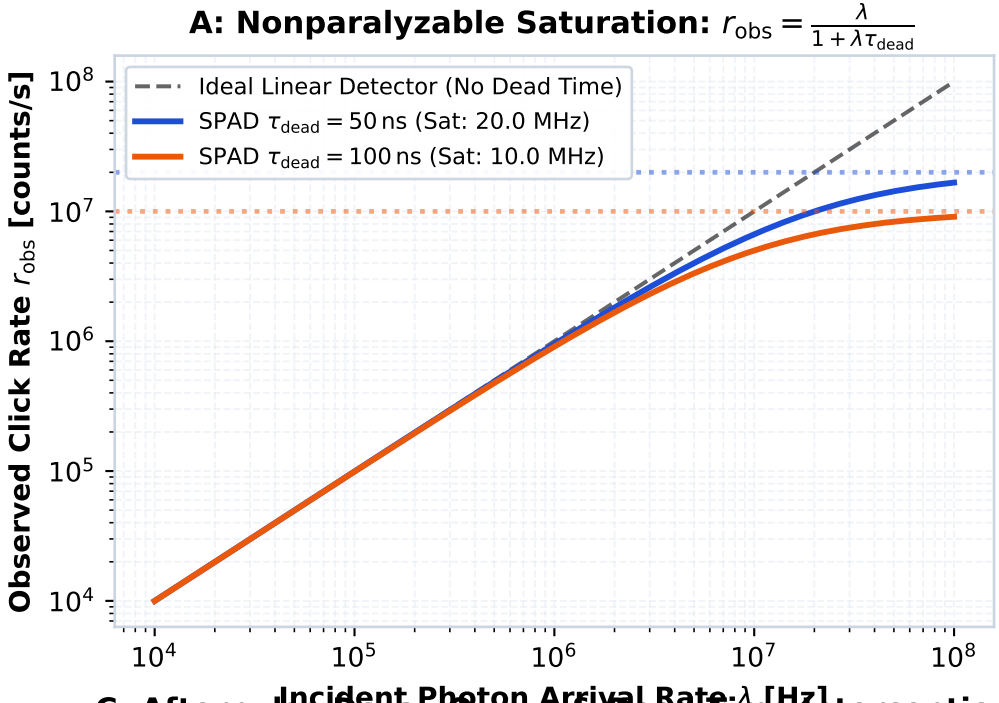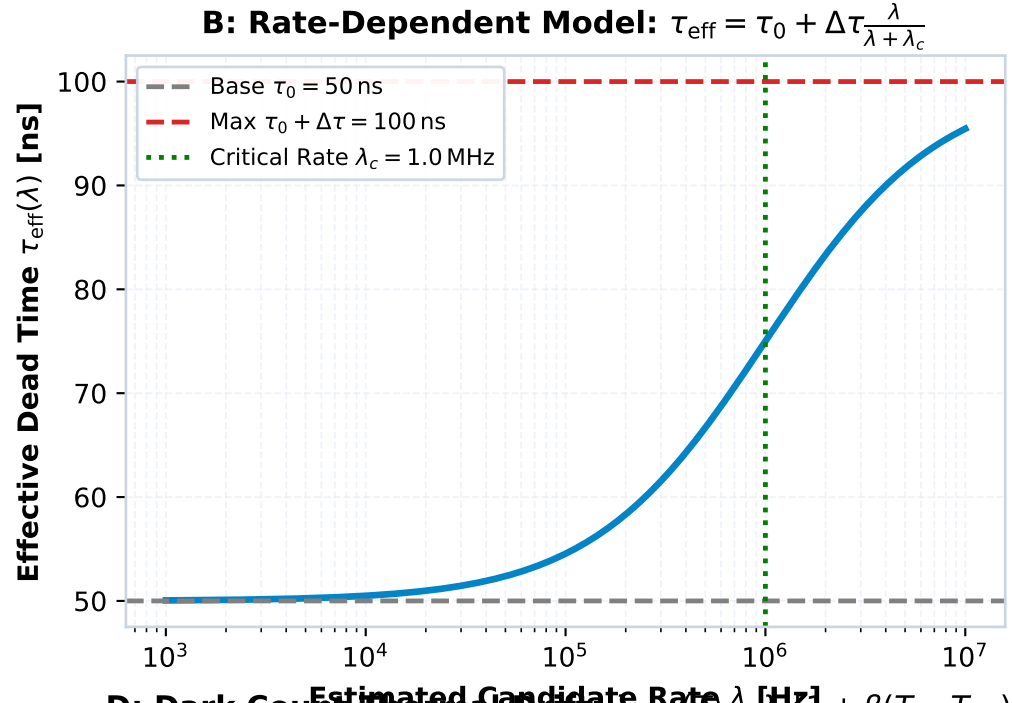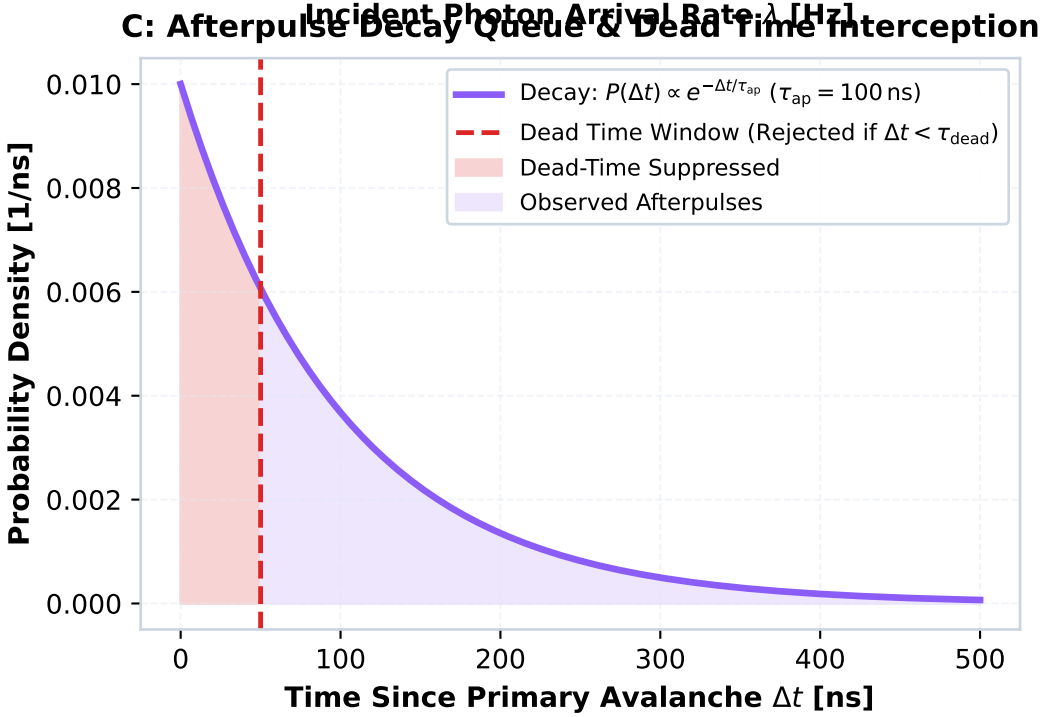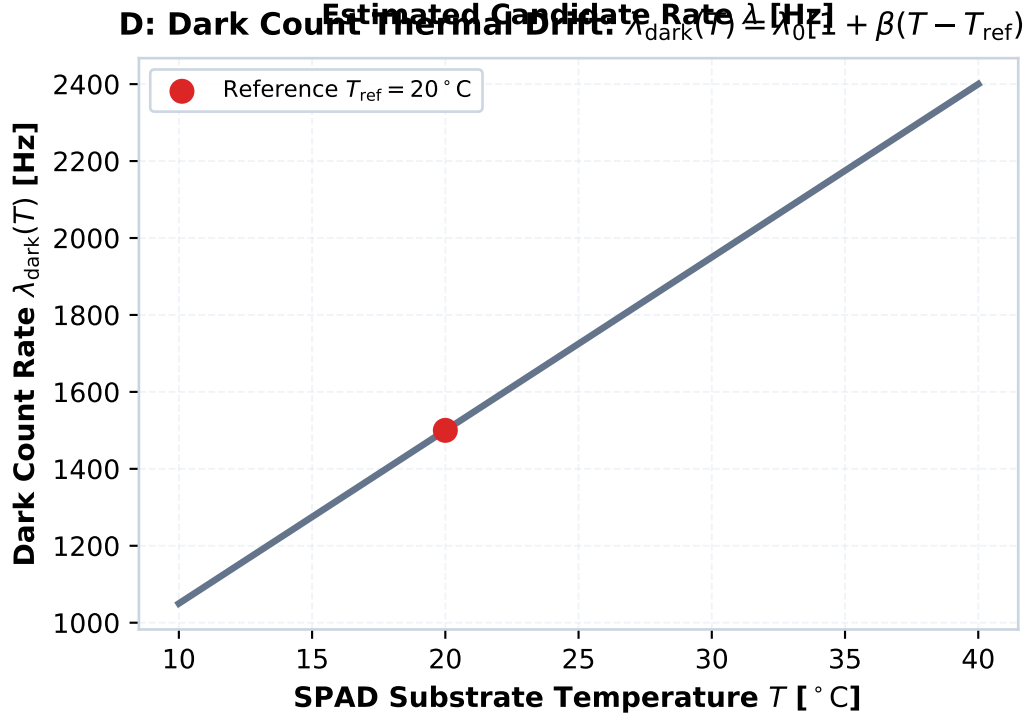Bob SPAD Detector Model: Dead Time Kinetics, Afterpulsing Dynamics & Thermal Noise